2 (Smoothing Splines, Textbook Ex. 5.7) Suppose $N \geq 2$, and that g is the natural cubic spline interpolant to the pairs $\{(x_i, z_i)\}_{i=1}^N$, with $-\infty < a < x_1 < \cdots < x_N < b < +\infty$, *i.e.* g as a cubic spline with knots $\{x_i\}_{i=1}^N$ satisfies

$$g(x_i) = z_i. \quad (1 \leq i \leq N)$$

Let $\widetilde{g} \in C^2[a, b]$ be any other twice continuously differentiable function supported on $[a, b]$ that interpolates the N pairs.

(a) Let $r = \widetilde{g} - g$ be the residual. Show that g'' is orthogonal to r'' in $L^2(\mathrm{d}\mathfrak{m})$ (denoting $\mathfrak{m}$ as the Lebesgue measure). That is,

$$\begin{aligned}\langle g'', r'' \rangle &:= \int_a^b g''(x) r''(x) \mathrm{d}x \\ &= -\sum_{j=1}^N g'''(x_j^+)[r(x_{j+1}) - r(x_j)] \quad \text{(by integration-by-part)} \\ &= 0,\end{aligned}$$

where $g'''(x_j^+)$ is the right third-order derivative of g at x_j.

Hint 1 *Note the stepwise constancies of g''' as a cubic spline function. Take care of the non-smoothness at knots in integration-by-part.*

(b) Show the Pythagorean identity

$$\|\widetilde{g}''\|_{L^2(\mathrm{d}\mathfrak{m})}^2 = \|g''\|_{L^2(\mathrm{d}\mathfrak{m})}^2 + \|r''\|_{L^2(\mathrm{d}\mathfrak{m})}^2 \geq \|g''\|_{L^2(\mathrm{d}\mathfrak{m})}^2,$$

and conclude that

$$\int_a^b \widetilde{g}''(x)^2 \mathrm{d}x \geq \int_a^b g''(x)^2 \mathrm{d}x,$$

with equality if and only if $g = \widetilde{g}$ *a.e.* $\mathfrak{m}$.

(c) Prove that the solution to the smoothing spline problem

$$\min_{f \in C^2[a,b]} \sum_{i=1}^N [y_i - f(x_i)]^2 + \lambda \int_a^b f''(x)^2 \mathrm{d}x \tag{1}$$

(for some $\lambda > 0$) must be a natural cubic spline with knots at $\{x_i\}_{i=1}^N$.

Hint 2 *The solution $\widehat{f}_\lambda$ might not interpolate $\{(x_i, y_i)\}_{i=1}^N$, but one might consider g interpolates $\{(x_i, \widehat{f}_\lambda(x_i))\}_{i=1}^N$.*

(a) $r=\tilde{g}-g$, $r(x_i)=0$, for g cubic spline, g''' is constant in $[x_j, x_{j+1}]$

$\Rightarrow \int_a^b g''r'' = \sum_{j=1}^{N-1}\int_{x_j}^{x_{j+1}} g''r''$, $\int_{x_j}^{x_{j+1}} g''r'' = g''r'\Big|_{x_j}^{x_{j+1}} - \int_{x_j}^{x_{j+1}} r'g''' = g''r'\Big|_{x_j}^{x_{j+1}} - g'''(x_j^+)\int_{x_j}^{x_{j+1}} r' = g''r'\Big|_{x_j}^{x_{j+1}} - g'''(x_j^+)r\Big|_{x_j}^{x_{j+1}}$

$\Rightarrow \int_a^b g''r'' = \sum_{j=1}^{N-1}\left(g''r'\Big|_{x_j}^{x_{j+1}} - g'''(x_j^+)(r(x_{j+1})-r(x_j))\right)$

$\sum_{j=1}^{N-1} g''r'\Big|_{x_j}^{x_{j+1}} = \sum_{j=1}^{N-1}(g''(x_{j+1})r'(x_{j+1}) - g''(x_j)r'(x_j)) = g''(x_N^-)r'(x_N) - g''(x_1^+)r'(x_1)$

g natural, so $g''(x_1)=g''(x_N)=0 \Rightarrow \sum_{j=1}^{N-1} g''r'\Big|_{x_j}^{x_{j+1}} = 0 \Rightarrow \int_a^b g''r'' = -\sum_{j=1}^{N-1} g'''(x_j^+)(r(x_{j+1})-r(x_j))$

for $r(x_i)=\tilde{g}(x_i)-g(x_i)=0$, $\int_a^b g''r'' = 0-0=0 = \langle g'', r''\rangle$

(b) $\tilde{g}=g+r \Rightarrow \tilde{g}''=g''+r'' \Rightarrow \|\tilde{g}''\|_{L^2}^2 = \int_a^b (g''+r'')^2 = \int_a^b (g'')^2 + \int_a^b (r'')^2 + \int_a^b 2g''r'' = \int_a^b (g'')^2 + \int_a^b (r'')^2$

$\Rightarrow \|\tilde{g}''\|_{L^2}^2 = \|g''\|_{L^2}^2 + \|r''\|_{L^2}^2 \Rightarrow \|\tilde{g}''\|_{L^2}^2 \geq \|g''\|_{L^2}^2 \Rightarrow \int_a^b \tilde{g}''(x)^2 dx \geq \int_a^b g''(x)^2 dx$

$\int_a^b \tilde{g}''(x)^2 dx = \int_a^b g''(x)^2 dx \Leftrightarrow \|r''\|_{L^2}^2 = 0 \Leftrightarrow r''=0$ a.e. in $[a,b] \Leftrightarrow r''\equiv 0$, for $r\in C^2[a,b] \Leftrightarrow r=\alpha+\beta x$

here $r(x_i)=0 \Rightarrow r=\alpha+\beta x \Leftrightarrow \beta=0, \alpha=0 \Leftrightarrow r\equiv 0 \Leftrightarrow \tilde{g}=g$

(c) Let $\hat{f}_\lambda$ be the best solution of $J(f)=\sum_{i=1}^{N}[y_i-f(x_i)]^2+\lambda\int_a^b f''(x)^2 dx$

Let g be natural cubic spline with knots at $(x_i, \hat{f}_\lambda(x_i))$

for all $h(x_i)=\hat{f}_\lambda(x_i)$, $h\in C^2[a,b]$, $\int_a^b g''^2 \leq \int_a^b h''^2$, let $h=\hat{f}_\lambda \Rightarrow \int_a^b g''^2 \leq \int_a^b \hat{f}_\lambda''^2$

for $g(x_i)=\hat{f}_\lambda(x_i) \Rightarrow \sum_{i=1}^{N}(y_i-g(x_i))^2 = \sum_{i=1}^{N}(y_i-\hat{f}_\lambda(x_i))^2 \Rightarrow J(g)\leq J(\hat{f}_\lambda)$

for $\hat{f}_\lambda$ is the best solution, $J(\hat{f}_\lambda)\leq J(g) \Rightarrow J(g)=J(\hat{f}_\lambda) \Rightarrow \hat{f}_\lambda$ is a natural cubic spline at $x_i\,{}_{i=1}^{N}$